METRIC

MIL-STD-2105E
6 January 2022

SUPERSEDING
MIL-STD-2105D
19 April 2011

DEPARTMENT OF DEFENSE TEST METHOD STANDARD

HAZARD ASSESSMENT TESTS FOR NON-NUCLEAR MUNITIONS



AMSC N/A AREA SAFT

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FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. The previous revision of this standard made a distinction between basic safety tests and the Insensitive Munition (IM) tests, even though these tests may often be contained in the same system hazard assessment test program. Additional or modified hazard testing may be required to fully assess the vulnerability of the given weapon system against the probable threats to which the system may be exposed. Accordingly, project managers and munition developers will conduct a Threat Hazard Assessment (THA) to determine the adequacy of safety or IM tests as referenced or specified in this standard. If the assessment indicates that different environmental hazards or threats to the weapon system pose additional vulnerability issues, supplement these tests to meet those additional requirements and provide rationale to support the assessment.

Section 3 Definitions have been updated to use the North Atlantic Treaty Organization (NATO) terminology for Allied Publications as listed in the NATO Terminology Database (NATOTerm). Section 4 General Requirements for IM tests have been updated to adopt requirements from United States (U.S)-ratified Allied Ordnance Publications (AOP). AOPs that have been updated since the issue of MIL-STD-2105D are munitions test procedures AOP-4240 (Fast Heating), AOP-4241 (Bullet Impact), AOP 4382 (Slow Heating), AOP-4396 (Sympathetic Reaction), AOP-4496 (Fragment Impact), and AOP-4526 (Shaped Charge Jet). Multiple figures have been added to section 4 to assist in communicating the information in that section.

These changes capture the Joint Service IM Technical Panel (JSIMTP) work to improve the weapon development and procurement process over the past 15 years by decreasing the overall burden of testing on programs while still providing sufficient data to fulfill the statutory requirement of identifying risk levels to the Office of the Secretary of Defense (OSD) and the Joint Staff. These improvements have impacted the process in the following ways:

a. IM/hazard classification (HC) harmonization – Eliminated a duplicative test approach by creating a single test strategy to satisfy the data requirements.

b. Cross-service IM harmonization – Office of the Under Secretary of Defense/Acquisitions, Technology, and Logistics (OUSD/AT&L) guidance in the “Joint Insensitive Munitions Test Standards and Compliance Assessment” memos provided clear direction for all services to use the same test procedures for each IM test, enabling full data sharing across the U.S. services and eliminating the need for re-testing for multi-service procurement.

c. IM score assessments – Replacement of full-scale IM testing with assessed IM scores where appropriate, as in the following cases:

(1) Modeling and simulation (M&S) used to determine the most effective test configuration for the Sympathetic Reaction (SR) test.

(2) M&S used to determine if a weapon will fail the SR or the Shaped Charge Jet (SCJ) Impact test. Therefore, a weapon can be assigned a failure without testing; saving money, time, and assets.

(3) Data and tests from similar systems, component level testing (engineering tests), and pre-production tests may be used to assign scores, especially when a weapon system upgrade only modifies electronic packages and not energetic systems.

(4) Existing explosive qualification test data and small-scale testing, when compared to IM test results, can reduce the number of tests required to assess a munition.

d. IM service board collaboration and procurement harmonization – Air Force, Army, and Navy IM service review organizations work closely together, which streamlines processes, identifies collaborative technologies, and accelerates cross-service procurement.

e. Streamlining and reducing planning and documentation – The Department of Defense (DoD) constantly improves the IM Strategic Plans (IMSP) Standard Operating Procedures (SOP). This makes each planning cycle easier for the Program Offices and Program Executive Office (PEO) to complete.

f. Reducing the out-of-cycle (OOC) IM waiver requirement and burden – The JSIMTP has worked with OUSD and the Joint Staff to remove the requirement to submit a waiver for minor weapons changes and has assumed the responsibility to prepare and present OOC waivers to the Joint Staff on behalf of the Program Offices.

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g. U.S. IM personnel joined international efforts to modify test standards – International working groups have modified IM test protocol documents to ensure that they reflect real world threat scenarios, most recently modifying the fast heating (FH) and slow heating (SH) test standards. Note that the FH test and the SH test were formerly called “Fast Cook-off” and “Slow Cook-off” tests in the previous revision.

h. Coordination of IM and hazard classification across NATO and other partner nations – This coordination ensures the safety of our warfighters in joint actions, helps U.S. services to easily procure and use foreign-produced munitions, and allows partner nations to purchase and use U.S.-produced systems in the most cost-effective manner.

3. Summary of Changes:

a. Figures: Removed all figures (figure 1, relationship of projection kinetic energy and hazard division (20 J); figure 2, “typical” item number and test sequence; and figure 3, “typical” spall impact test configuration); added figure 1, typical basic safety tests; figure 2, DoD IM test plan development and approval process; figure 3, standard IM test plan; and figure 4, harmonized IM/HC test requirements.

b. Tables: Added table I, terms and references; table II, standard IM tests; table III, units of measurement and abbreviations; and table IV, implemented international standardization agreements.

c. Foreword: Added in depth overview of changes from MIL-STD-2105D, updated guidance on munition test modification, and updated point of contact for comments, suggestions, or questions on this document.

d. Section 2: Updated to include all references specified in sections 3, 4, or 5 of this standard in accordance with MIL-STD-962.

e. Section 3 Definitions: Removed all definitions and provided reference documents for applicable definitions. Added table of acronyms. Added the following terms: blast overpressure, acceptor munition, donor munition, explosive component, hazard classification, hazard division, logistical configuration, and tactical configuration. Removed the following terms: all-up-round, bare round or configuration, explosive device, exudation, hazardous fragment, munition subsystem, propulsion, service review organization, and threat hazard assessment.

f. Section 4 General Requirements: Improved layout of the purpose and sequence of basic safety test, THA, standard IM test plan, tailored IM test plan, harmonized IM/HC test plan, and IM test report.

g. Section 5 Detailed Requirements: Changed “12-meter (40-foot) drop test” to “safety drop test” throughout this document. Changed “radiographic inspection” paragraphs to “radiographic/visual inspection” under 28-day T&H test, vibration test, and 4-day T&H test. Changed “radiographic inspection” paragraph to “radiographic inspection (optional)” under safety drop test. Removed the following paragraphs: additional test (the use of additional tests is now addressed in 4.1.1) and spall impact test; updated the following paragraphs: instrumentation, test assessment and analysis report, IM tests, FCO, SCO, bullet impact (BI), fragment impact (FI), SR, and SCJ.

h. Section 6 Notes: Removed data item descriptions references table and the tailoring guidance paragraph. Updated the following paragraphs: service review organizations, tests for HC, units of measurement and abbreviations, international standardization agreement implementation, subject term (key word) listing.

4. This document contains a description of tests or references to NATO AOP and STANAGs for the assessment of munition safety and IM characteristics of non-nuclear munitions. Historically, this standard was primarily used for the assessment of weapon safety. The standard was later revised to add additional IM tests as called out by the Joint Service Requirement for Insensitive Munitions (JSRIM) and now by various NATO STANAGs/AOPs.

5. Three sets of tests are commonly used to assess munitions with respect to hazards: IM tests as contained in this standard, HC tests used to classify munitions for transportation and storage purposes, and basic safety tests used to assess munition safety and suitability or system vulnerability. In order to best utilize limited resources and avoid test redundancy, coordinate IM test plans to the maximum extent possible, so that all three sets of tests can be addressed in one coordinated test program with the minimum number of Test Items. It is recommended that test plans be coordinated with the appropriate service hazard classifier and approved by the Department of Defense Explosives Safety Board (DDESB), 4800 Mark Center Drive, Suite 16E12, Alexandria, VA 22350, when a DoD HC for an item is to be obtained in accordance with TB 700-2.

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6. Program Managers (PM) are responsible for planning and executing a hazard assessment test program. The program includes a test plan based on a realistic life cycle environmental profile (LCEP). Prior to testing, the PMs should establish safety design goals for the test plan and have these goals approved by the service review organization within the applicable department. Following testing, the PMs should submit to those same service review organizations all test reports and available data (i.e., engineering or analogous hardware test, small scale test, gap test, or M&S test results) that may be useful for determining munitions' responses to IM testing.

7. The service review organization should review the test plan and test report and examine the results of the hazard assessment testing and assessment scores to ensure that safety and IM requirements are met. The service review organization should produce a final recommendation for or against the service use of the weapon system. If cross-service use is likely, full data sharing and review will eliminate the need for re-evaluation for other service use. For joint programs, all affected service review organizations should conduct this review and examination and develop a final recommendation.

8. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, D.C. 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This standard provides test procedures for the conduct of basic safety and IM tests for all DoD non-nuclear munitions, munition subsystems, and explosive devices.

1.2 Purpose. A framework for the development of a consolidated safety and IM assessment test program for non-nuclear munitions is provided. The tests are to characterize the munitions and provide the service review organization the information required to assess the safety and IM compliance of non-nuclear munitions.

1.3 Application. This standard applies to all non-nuclear munitions (e.g., missiles, rockets, artillery ammunition, bombs) and munitions subsystems (e.g., warheads, fuzes, cartridge actuated devices, propulsion units, safe and arm devices, pyrotechnic devices, chemical payloads) and other explosive devices.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL STANDARDIZATION AGREEMENTS

AAS3P-01	-	Safety and Suitability for Service Assessment Testing of Non-Nuclear Munitions
AOP-39	-	Policy for Introduction and Assessment of Insensitive Munitions (IM)
AOP-39.1	-	Guidance on the Organization, Conduct and Reporting of Full Scale Tests (SRD)
AOP-4240	-	Fast Heating Munition Test Procedures
AOP-4241	-	Bullet Impact Munition Test Procedures
AOP-4382	-	Slow Heating Test Procedures for Munitions
AOP-4396	-	Sympathetic Reaction Test Procedures for Munitions
AOP-4496	-	Fragment Impact Test Procedures for Munitions
AOP-4526	-	Shaped Charge Jet Munition Test Procedures
STANAG-4375	-	Safety Drop, Munition Test Procedures

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-331	-	Fuzes, Ignition Safety Devices and Other Related Components, Environmental and Performance Tests for
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-882	-	System Safety

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JOINT ORDNANCE TEST PROCEDURE

JOTP-001 - Allied Ammunition Safety and Suitability for Service Assessment Testing Publication - Guidance

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

TECHNICAL BULLETINS

TB 700-2 - Department of Defense Ammunition and Explosives Hazard Classification Procedures

(Copies of this document are available online at https://liw.logsa.army.mil/etmapp/#/etm/search.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Terms. The terms used in this standard and their source reference are provided in table I.

TABLE I. Terms and references.

Term	Reference
Blast Overpressure	NATOTerm[1/]
Acceptor Munition	NATOTerm
Donor Munition	NATOTerm
Explosive	NATOTerm
Explosive Component	NATOterm
Hazard Classification	TB 700-2
Hazard Division	TB 700-2
Insensitive Munition	NATOTerm
Logistical Configuration	SRD AOP-39.1
Munition	NATOterm
Response (Reaction) descriptors - Detonation (Type I), Partial Detonation (Type II),Explosion (Type III), Deflagration (Type IV), Burn (Type V) & No Reaction (Type VI)	AOP-39
Sympathetic Reaction	NATOTerm
Tactical Configuration	SRD AOP-39.1
Weapon System	NATOTerm

NOTE:
1/ The NATO terminology database is found at https://nso.nato.int/natoterm/.

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3.2 Acronyms used in this standard. The acronyms used in this standard are defined as follows:

AOP	Allied Ordnance Publication
AUR	All-Up-Round
BI	Bullet Impact
DDESB	Department of Defense Explosives Safety Board
DoD	Department of Defense
EOD	Explosive Ordnance Disposal
FCO	Fast Cook-Off
FH	Fast Heating
FI	Fragment Impact
HC	Hazard Classification
IM	Insensitive Munition
IMHCO	IM Hazard Classification Office
IMSP	IM Strategic Plan
JSIMTP	Joint Service IM Technical Panel
JSRIM	Joint Service Requirement for Insensitive Munitions
LCEP	Life Cycle Environmental Profile
M&S	Modeling and Simulation
NATO	North Atlantic Treaty Organization
NATOTerm	NATO Terminology Database
OOC	Out of Cycle
OSD	Office of the Secretary of Defense
OUSD/AT&L	Office of the Under Secretary of Defense/Acquisitions, Technology, and Logistics
PEO	Program Executive Office
PM	Program Manager
SCJ	Shaped Charge Jet
SCO	Slow Cook-Off
SH	Slow Heating
SOP	Standard Operating Procedure
SR	Sympathetic Reaction
STANAG	Standardization Agreement
T&H	Temperature and Humidity
THA	Threat Hazard Assessment
U.S.	United States

4. GENERAL REQUIREMENTS

4.1 Basic safety tests. A test plan shall be prepared based on the munition LCEP. Requirements for an LCEP are specified in MIL-STD-882, MIL-STD-810, AAS3P-01, and Joint Ordnance Test Procedure (JOTP)-001. A typical basic safety test sequence includes 28-day temperature and humidity (T&H) cycling, vibration (transportation, aircraft and shipboard) testing, 4-day T&H cycling, and safety drop testing as illustrated on figure 1.

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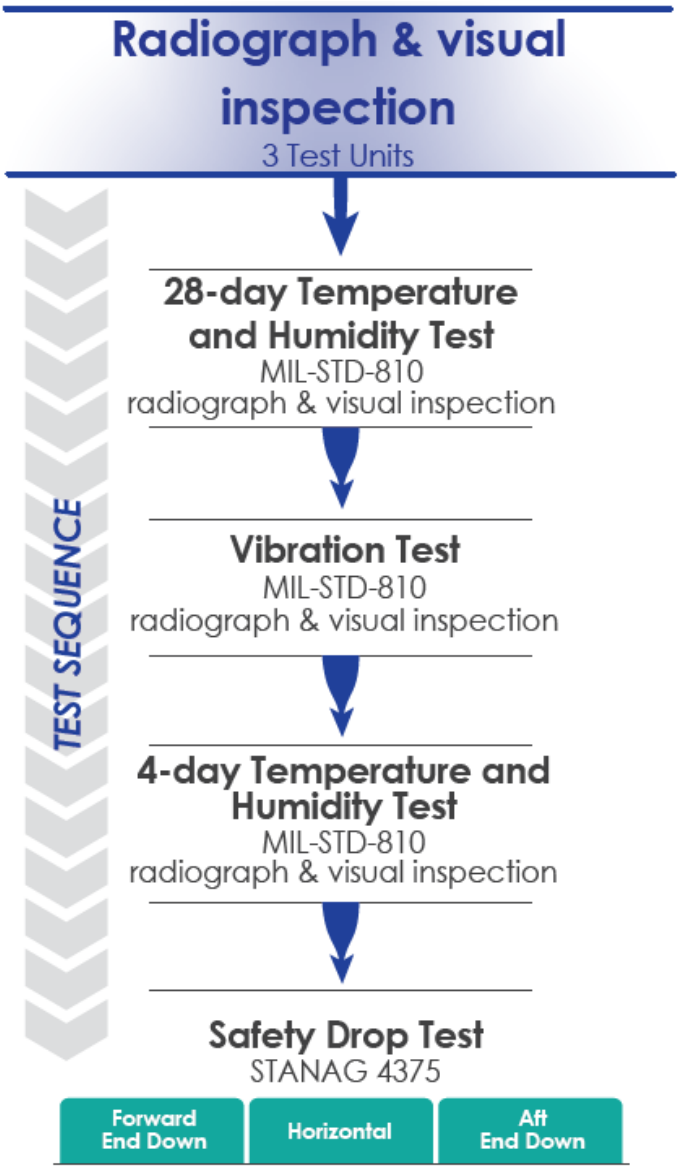


FIGURE 1. Typical basic safety tests.

4.1.1 Basic safety test plan. A basic safety test plan shall be developed by the PM based on the LCEP. The test plan shall include provisions for the conduct and sequence of tests and required environmental conditioning. The test plan shall be reviewed and concurred with by the appropriate service review organization(s). The service review organization(s) may recommend or authorize variations to the tests and procedures in section 5. The test plan may include additional tests selected from other sources or devised to investigate hazardous conditions and environments identified by hazard analyses performed as part of the system safety program specified in MIL-STD-882. Data collection requirements are provided in AOP-39.

4.1.2 Basic safety test report. The data gathered as required in section 5 shall be used to prepare a test report that summarizes the results and conclusions in accordance with AAS3P-01 and JOTP-001. The test report shall be reviewed by the service review organization(s).

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4.2 IM tests. An IM THA that identifies the threats that a munition may be exposed to during its lifecycle shall be developed. The THA will serve as a guide for the development of an IM/HC test plan. Guidance from the service technical review boards/panels (service review organization) shall be incorporated into the determination of the test methods and configurations (i.e., standard, tailored, or harmonized IM/HC) developed as illustrated on figure 2.

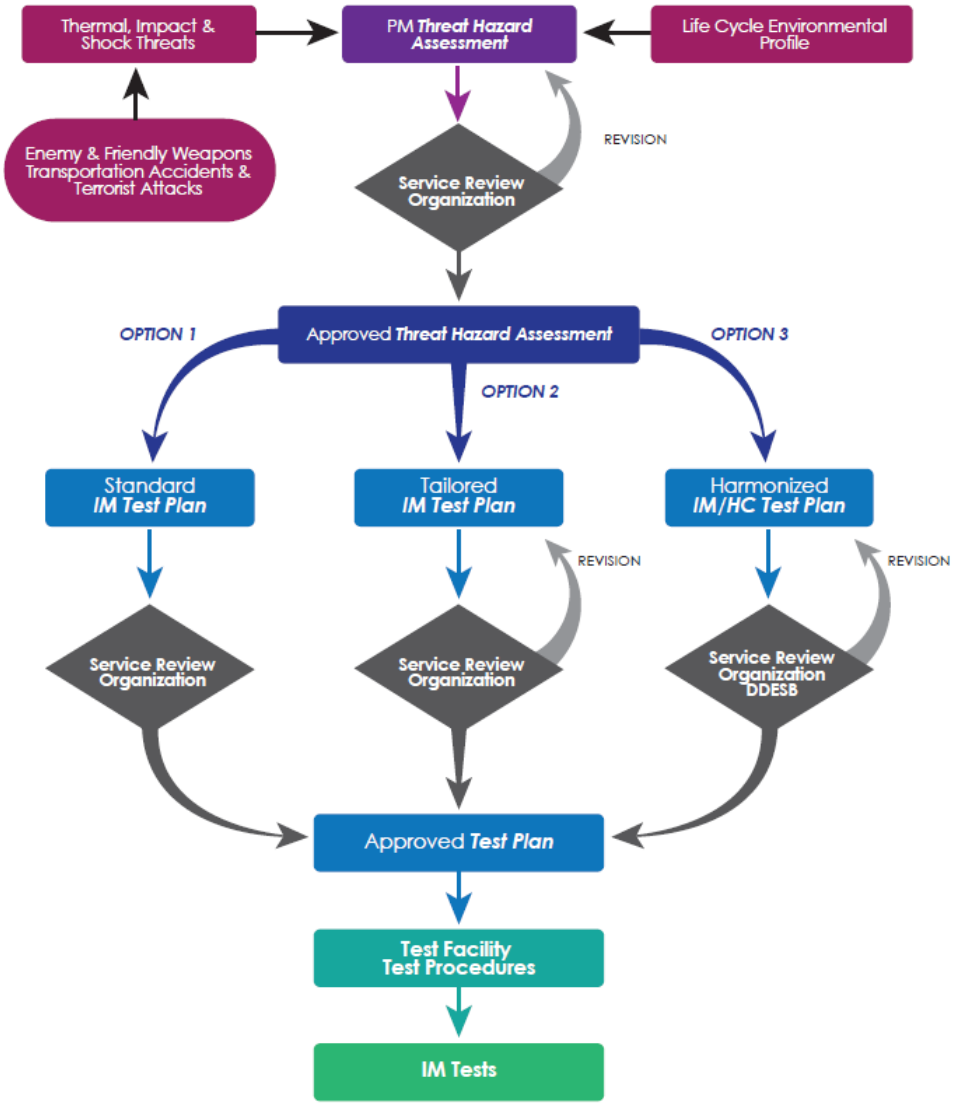


FIGURE 2. DoD IM test plan development and approval process.

4.2.1 THA. AOP-39 and the munition LCEP shall be used to determine the thermal, shock, and impact threats in which the munition may be exposed. The assessment shall include threats posed by friendly munitions, potential enemy munitions, accidents and mishaps, handling, and terrorist attacks. The THA shall be based on analytical or empirical data for the purposes of identifying the worst-case threat scenarios, munition configuration(s), and the anticipated munition response to each IM stimuli using the requirements of AOP-39. The THA shall be submitted to the appropriate service review organization(s) for review and approval.

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4.2.2 Standard IM test plan. If the THA verifies that the standard IM stimuli are appropriate and the service review organization(s) approve, the standard IM test series shall be conducted. A test plan using AOP-39 shall be prepared and approved by the service review organization(s). Table II and figure 3 provide a summary of the test requirements. Test facilities shall provide the instrumentation and range required to obtain the data required to determine the Test Item type of reaction (e.g., type I through type VI per AOP-39). Details of instrumentation requirements (e.g., blast overpressure, photographic documentation, witness plates) are provided in each NATO test procedure listed in table II.

TABLE II. Standard IM tests.

Test	NATO Procedure	No. Tests	Method
FH	AOP-4240	2	Method 1 or Method 2
SH	AOP-4382	2	Method 1
BI	AOP-4241	2	Method 1
FI	AOP-4496	2	Method 1
SR	AOP-4396	2	
SCJ	AOP-4526	2	Method 1

If the logistical and tactical configurations are identical, two FH, two BI, two FI, and two SCJ tests are required in that configuration. Two SR tests are required in the logistical configuration. The number of munitions tested in SR shall be defined by the shipping configuration (e.g., unit load or pallet). SH is a storage only test and is not conducted in the tactical configuration. Requirements are provided in AOP-39.

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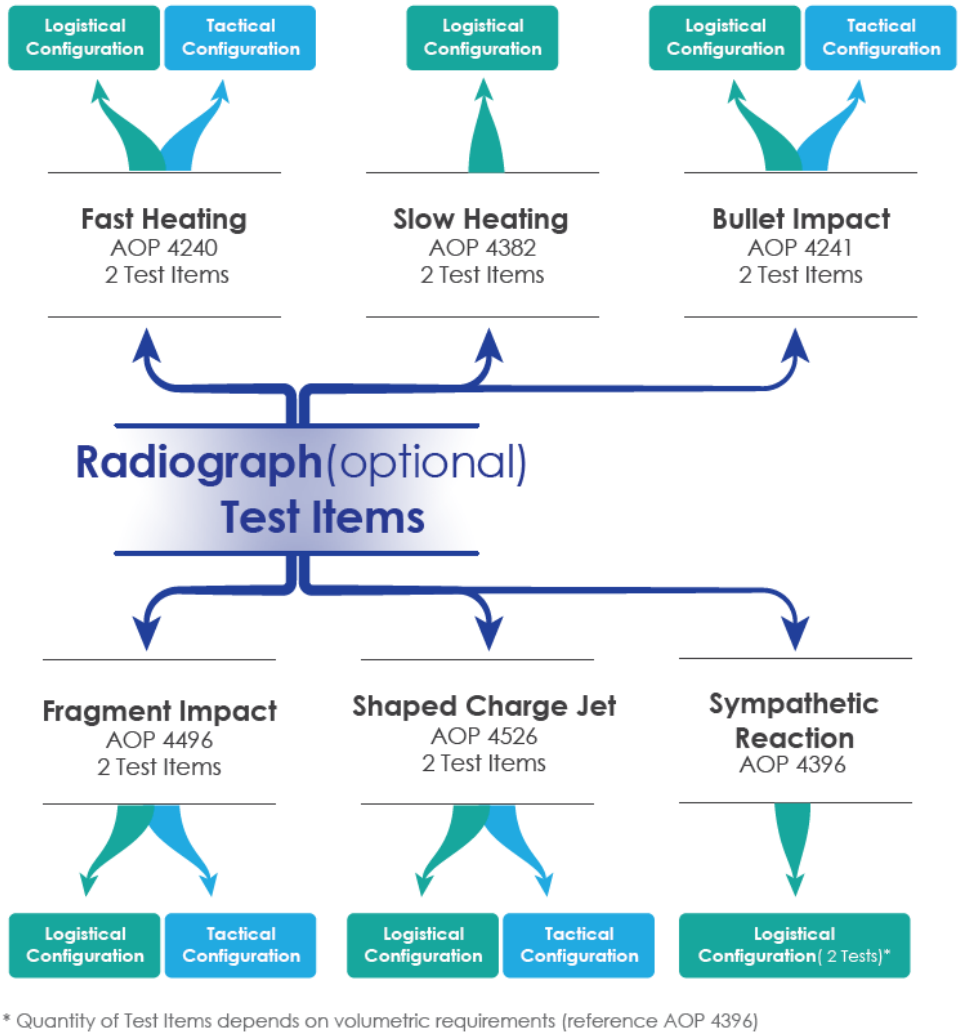


FIGURE 3. Standard IM test plan.

4.2.3 Tailored IM test plan. Should the approved THA indicate that the standard IM stimuli are either not appropriate or not applicable, develop a tailored IM test plan using in AOP-39 for approval by the service review organization(s). The plan shall provide the threat characteristics and Test Item configuration(s) for each identified IM stimulus. One test in the logistical configuration and one test in the tactical configuration are required for each identified IM stimulus. If the logistical and tactical configurations are identical, two tests in this configuration, for each tailored IM threat stimuli, shall be conducted. The tailored IM test plan may include the elimination of IM test(s) if the munition is expected to fail to meet the test passing criteria of AOP-39 and the service review organization(s) approve. Predictions developed from M&S efforts may also be used to eliminate full-scale tests if approved by the service review organization(s).

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4.2.4 Harmonized IM/HC test plan. The PM may prepare a combined IM/HC test plan by harmonizing FH and SR tests with DDESB HC requirements as shown on figure 4. The FH and SR (confined and unconfined) tests shall be tested in the shipping configuration (i.e., palletized or unit load). Munitions that meet most IM requirements when tested in accordance with an approved harmonized IM/HC test plan may be assigned to either HD 1.2.3 (Non-Mass Explosion, Fragment Producing) or HD 1.6 (Explosive Article, Extremely Insensitive, No Mass Explosion Hazard) using the procedures specified in TB 700-2. The IM/HC plan shall be coordinated with the service review organization(s), the Service Hazard Classifier, and the DDESB. The plan shall include complete descriptions of Test Item configuration for each test and data measurement requirements. Requirements are provided in AOP-39.

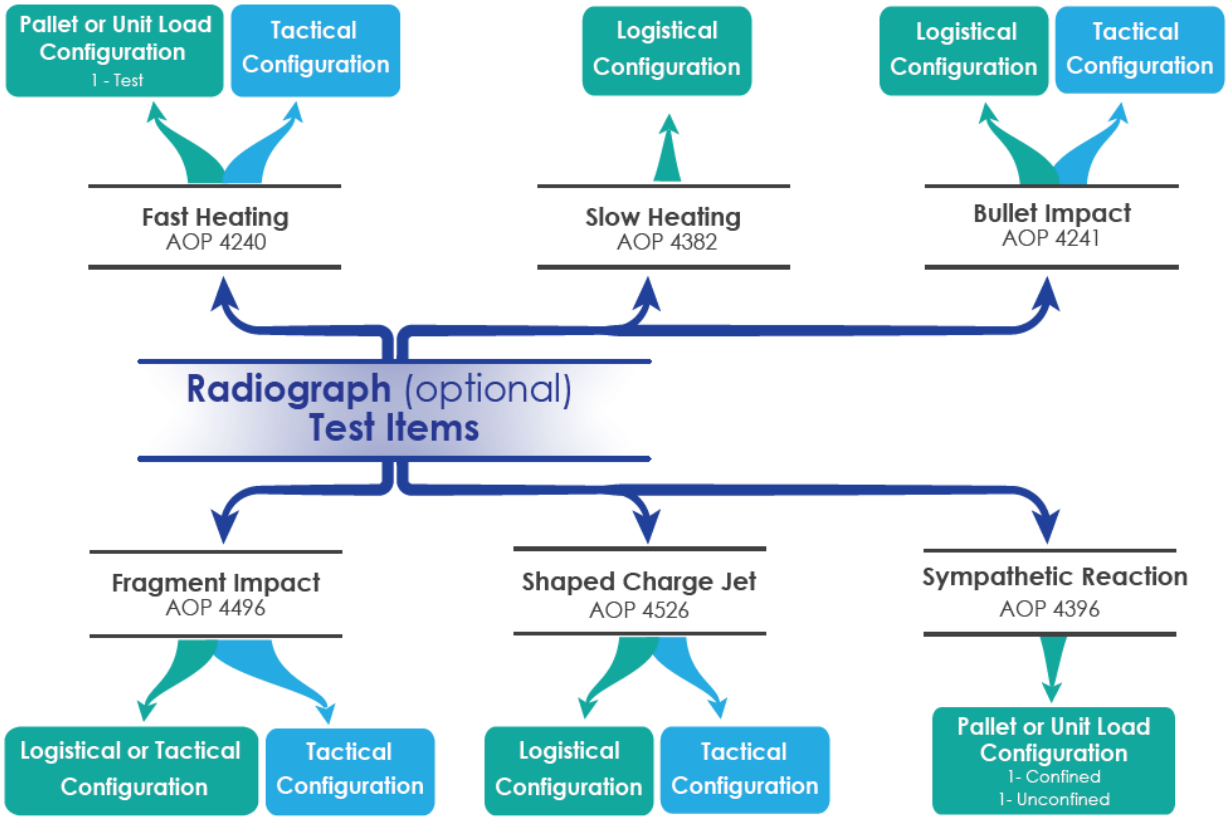


FIGURE 4. Harmonized IM/HC test requirements.

4.3 IM test report. A detailed IM test report shall be prepared that documents the results of the tests. The report shall contain detailed data specified in section 5 for each test to allow service review organizations to determine the reaction level (type I through type VI). Detailed data reporting requirements are provided in AOP-39.

5. DETAILED REQUIREMENTS

5.1 Basic safety tests. The following tests shall be considered for inclusion in the hazard assessment test program. Three Test Items shall be subjected sequentially to the following tests as shown on figure 1.

5.1.1 28-day T&H test. The 28-day T&H test shall be as follows.

5.1.1.1 Description of test. The 28-day T&H test consists of exposing the Test Item to alternating 24-hour periods (no period lasting less than 24 hours) of high and low temperatures for a total of 28 days. Derive the temperature range and relative humidity from the LCEP (see 4.1). Test a minimum of three Test Items.

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5.1.1.2 Test procedure. Develop test procedures (refer to AAS3P-01 and JOTP-001) that reflect the temperature and humidity conditions documented in the LCEP. Visually examine each Test Item prior to testing and record the appropriate critical dimensions. Radiographically examine the Test Items to determine material condition prior to the test.

5.1.1.2.1 Test facilities. Use chambers capable of producing the required temperatures and humidity over the time spans specified in 5.1.1.1 that do not obstruct the free flow of air in contact with the item under test. Separate chambers for each test environment specified are required.

5.1.1.2.2 Temperature cycling. Begin the test by subjecting the Test Item to either the high or low temperature environment for a 24-hour period. At the end of this period, transfer the Test Item to the other environment. The transfer should be accomplished in less than 30 minutes and the actual time should be recorded in the test data report (refer to AAS3P-01 and JOTP-001). At the end of each high and low temperature cycle change, inspect the Test Item for damage and collect any exudate for chemical analysis. Continue testing and inspecting for the number of periods specified.

5.1.1.2.3 Test interruptions. Minimize interruptions of the test. If the test is interrupted by slack labor periods (weekends, holidays), maintain the last test environment encountered prior to the slack period. Extend the test period as necessary to complete at least 20 temperature changes (hot/cold) or 10 full cycles. A full cycle consists of two temperature changes (e.g., hot-to-cold-to-hot).

5.1.1.3 Instrumentation. Continuously monitor and record the temperature and humidity levels of the test chambers.

5.1.1.3.1 Photography. Photographs of the Test Items shall be taken to record the condition of the Test Items and test setup prior to and after the test (refer to AAS3P-01 and JOTP-001).

5.1.1.3.2 Radiographic/visual inspection. Conduct a radiographic and visual inspection in accordance with ASTM E1742/E1742M to record the condition of the Test Item prior to and after the test.

5.1.1.4 Passing criteria. The following passing criteria are based on the final observation:

a. No reaction of the explosive.

b. No exudation containing explosive material.

c. Explosives do not crack or separate in a manner that would create a hazardous condition.

d. All safety devices remain in the safe position or safe condition.

e. The structural integrity of the item is not compromised by corrosion, loosening of joints, or other physical distortions.

5.1.1.5 Documentation. Collect the data required by AAS3P-01 and JOTP-001.

5.1.2 Vibration test. The vibration test shall be as follows.

5.1.2.1 Description of test. The vibration test consists of exposing the Test Item to the most intense vibration environment that it will encounter during the life cycle as determined by the LCEP. Test a minimum of three items that have undergone testing in accordance with 5.1.1.

5.1.2.2 Vibration orientation. Conduct vibration tests along the appropriate mutually perpendicular axes.

5.1.2.3 Vibration schedule. Determine the vibration schedule from the LCEP (see 4.1).

5.1.2.3.1 Changes in vibration schedule. Changes in the selected schedule of vibration levels, frequency ranges, and time duration of the test can be selected by the PM or the procuring activity with the approval of the service review organization.

5.1.2.3.2 Test temperatures. Conduct vibration tests at low and elevated temperatures or ambient temperature identified by the LCEP.

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5.1.2.4 Test procedures. Develop test procedures that reflect vibration modes and temperatures anticipated in the LCEP. Consider vibration environments as specified in MIL-STD-810 including one or more of the following.

5.1.2.4.1 Transportation vibration. Vibrate the Test Item in the normal configuration as shipped (i.e., in a shipping container, if used). The Test Item may be vibrated in the bare configuration if it can be shown that testing in the bare configuration produces an equivalent environment. The Test Item shipped and stored in a ready service configuration should be vibrated in a fixture and orientation representative of that configuration.

5.1.2.4.2 Aircraft vibration. Vibrate the item in the configuration utilized for aircraft transportation (fixed wing and rotary wing) or combat carriage.

5.1.2.4.3 Shipboard vibration. Vibrate the item in its shipboard stowage configuration. Should the item be carried on a launcher or in a ready service configuration, vibrate the item in a fixture and orientation representative of that configuration.

5.1.2.5 Instrumentation. Record the test equipment inputs (acceleration, power spectral density, etc.) and Test Item responses from accelerometers installed on the Test Item and vibration equipment. Record Test Item temperature as required by the test plan.

5.1.2.5.1 Photography. Use photographs to record the Test Item configuration and installation on the vibration equipment prior to the test and the Test Item condition after the test (refer to AAS3P-01 and JOTP-001).

5.1.2.5.2 Radiographic/visual inspection. Conduct a radiographic and visual inspection in accordance with ASTM E1742/E1742M to record the condition of the Test Item prior to and after the test.

5.1.2.6 Passing criteria. The following passing criteria are based on the final observation:

a. No reaction of the explosive.

b. No exudation containing explosive material.

c. Explosives or propellants do not crack or separate in a manner that would create a hazardous condition.

d. All safety devices remain in the safe condition.

e. The structural integrity of the Test Item is not compromised by corrosion, loosening of joints, or other physical distortions.

5.1.2.7 Documentation. Collect the data required by AAS3P-01 and JOTP-001.

5.1.3 4-day T&H test. The 4-day T&H test shall be as follows.

5.1.3.1 Description of test. The 4-day T&H test is a 4-day version of the 28-day T&H test and consists of exposing the item to alternating 24-hour periods of temperature and relative humidity as derived from the LCEP (see 4.1). The same data as the 28-day T&H test is required for the 4-day T&H test (see 5.1.1, AAS3P-01, and JOTP-001). The three Test Items that have undergone testing in accordance with 5.1.1 and 5.1.2 shall be subjected to two complete 24-hour cycles at each temperature.

5.1.3.2 Radiographic/visual inspection. Conduct a radiographic and visual inspection in accordance with ASTM E1742/E1742M to record the condition of the Test Item prior to and after the test.

5.1.3.3 Passing criteria. The following passing criteria are based on the final observation:

a. No reaction of the explosive.

b. No exudation containing explosive material.

c. Explosives or propellants do not crack or separate in a manner that would create a hazardous condition.

d. All safety devices remain in the safe condition.

e. The structural integrity of the Test Item is not compromised by corrosion, loosening of joints, or other physical distortions.

5.1.3.4 Documentation. Collect the data required by AAS3P-01 and JOTP-001.

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5.1.4 Safety drop test. The safety drop test shall be as follows.

5.1.4.1 Test procedures. Perform this test in accordance with STANAG-4375. The safety drop height shall be derived from the LCEP and approved by the service review organization. However, the minimum drop height shall be no less than 12 meters (40 feet). Additional drop requirements for fuzes and initiation devices are specified in MIL-STD-331.

5.1.4.2 Radiographic inspection (optional). A radiographic inspection in accordance with ASTM E1742/E1742M to record the condition of the Test Item prior to and after (if safe) the test is optional.

5.1.4.3 Test Item configuration. Unless approved by the service review organization(s), the Test Item shall be dropped in the configuration(s) specified in STANAG-4375. The Test Item shall consist of a production-representative arrangement, although non-explosive munition components, assemblies, sections, or units need to only be geometrically (including weight and balance) representative.

5.1.4.4 Test surface. The impact surface shall be as specified in STANAG-4375. The impact surface and recovery area shall be positioned and configured to facilitate safe handling and disposal of dropped munitions, and to afford appropriate protectiveness to exposed assets (e.g., facilities, people) if a safety drop trial results in an explosive reaction (i.e., a test failure).

5.1.4.5 Orientation and number of drops. A minimum of three separate drops at different impact orientations shall be conducted. Additional drops may be conducted, as determined by the LCEP or leading service's requirements. The Test Item is to be released (dropped) within two degrees off-axis. The initial impact is in the following minimum orientations:

a. Major axis vertical, nose down ↓

b. Major axis vertical, aft end down ↑

c. Major axis horizontal, →

Drop tests shall be carried out in order of decreasing likelihood of producing unwanted events. No Test Item shall be dropped more than once.

5.1.4.6 Conditioning and pre-stressing. For the purpose of this document, conditioning is defined as the steady-state temperature an item achieves and maintains throughout the test procedure, typically -54 °C (-65 °F) and +71 °C (+160 °F).

Pre-stressing is defined as those specific environmental effects (temperature, humidity, vibration) of 4.1.1 or the munition type-specific AOP, which comprise the immediate history of the Test Item just prior to the drop.

The following apply to the safety drop test:

a. Category 1. Missiles, large rockets (not man-carried), and tactical air released munitions will be dropped at ambient temperature and will have previously undergone environmental pre-stressing in accordance with 4.1.1 or the munition type-specific AOP. The Test Item may be dropped while conditioned to extreme temperatures to satisfy other requirements.

b. Category 2. All munitions that are not category 1 shall be subjected to conditioning immediately prior to the test to simulate the likely worst-case temperatures at the moment of a free fall in the service environment. The environments to which the Test Item should be conditioned/pre-stressed shall be determined by the responsible acquisition authority based on the LCEP or munition type-specific AOP. The Test Items may be pre-stressed to satisfy other requirements.

For items in category 2, the LCEP shall determine the temperature extreme, to which the Test Item is conditioned. The extreme temperature, typically -54 °C (-65 °F) and +71 °C (+160 °F), should be selected so as to represent the hot or cold temperature at which historical data on similar items has proved to be most sensitive to impact. Any other selected temperatures shall be justified in the safety data package and approved by the lead service review organization.

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5.1.4.7 Passing criteria. The following passing criteria are based on the final observation:

a. No visible or audible reaction of the propellant/pyrotechnic/explosive in the Test Item.

b. The Test Item remains Safe for Disposal. An item is Safe for Disposal if:

(1) The munition is unusable and shall maintain its safety features including Explosive Ordnance Disposal (EOD) features in a condition that will permit its disposal without injury to personnel using the applicable handling and disposal regulations and procedures.

(2) No unintended burning, detonation, or spread of propellant/pyrotechnic/explosive or other materials that present a significant hazard shall occur as a result of the event or during the physical removal and disposal of a damaged munition.

(3) The munition shall remain sufficiently safe to allow movement to a suitable location for disposal.

5.1.4.8 Test assessment and analysis report. Collect the data required by AAS3P-01 and JOTP-001. (Note: The safety drop test results may be used for HC purposes if the configuration is harmonized with TB 700-2 and approved by the DDESB.)

5.2 IM tests . The tests contained in or referenced in this section provide the basis to test munitions against credible potential thermal, shock, and impact threats and evaluate the ability of the munition to meet passing criterion. The service review organization(s) may assess the response of a munition using M&S predictions, similarly to other munitions or previous experience in lieu of full-scale all-up-round (AUR) tests.

5.2.1 FH test. This test shall be performed in accordance with AOP-4240, method 1 (large pool fire) or method 2 (fuel burner fire). Method 2 provides financial (i.e., reduced fuel costs) and environmental advantages (i.e., greatly reduced particulate and hydrocarbon emissions) as compared to method 1 and should be used, if possible. Two tests shall be conducted, one each in the logistical and tactical configuration. Data collection requirements and passing criteria are provided in AOP-39.

5.2.2 SH test. This test shall be performed in accordance with AOP-4382, method 1 (standard test). Two tests shall be conducted in the logistical configuration. Data collection requirements and passing criteria are provided in AOP-39.

5.2.3 BI test. This test shall be performed in accordance with AOP-4241, method 1 (standard test). Two tests shall be conducted, one each in the logistical and tactical configuration. The aimpoint for the bullets shall be determined by the THA and approved by the service review organization(s) and, as appropriate, by the DDESB. Additional guidance for aimpoints and shotlines is provided in AOP-4241. Data collection requirements and passing criteria are provided in AOP-39.

5.2.4 FI test. This test shall be performed in accordance with AOP-4496, method 1 (standard test). Two tests shall be conducted, one each in the logistical and tactical configuration. The aimpoint for the fragment shall be determined by the THA and approved by the service review organization(s) and, as appropriate, by the DDESB. Data collection requirements and passing criteria are provided in AOP-39.

5.2.5 SR test. This test shall be performed in accordance with AOP-4396. Two tests shall be conducted in the logistical configuration. The test item configuration will be determined by the THA and approved by the service review organization(s) and, as appropriate, by the DDESB. External confinement shall be used for one of the two tests unless it is stated otherwise in a combined IM/HC test plan that was approved by the service review organization(s) and by the DDESB. Data collection requirements and passing criteria are provided in AOP-39.

5.2.6 SCJ test. This test shall be performed in accordance with AOP-4526, method 1. Two tests shall be conducted, one each in the logistical and tactical configuration. An 81-millimeter LX-14 loaded precision shaped charge shall be fired into the Test Item along the shotline(s) identified as being the worst case(s) in the THA, which was approved by the service review organization(s). Data collection requirements and passing criteria are provided in AOP-39.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The tests described herein or referenced are used to assess the safety and IM characteristics of non-nuclear ordnance. The ordnance covered by these tests is designed for military use only; thus, this standard has no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the title, number, and date of this standard.

6.3 Submission of test reports and results. Submit copies of test reports and results to the following address for storage in the IM Hazard Classification Office (IMHCO):

Navy/Marine Corps - (for IM):

Commanding Officer
Naval Ordnance Safety & Security Activity Farragut Hall
ATTN: NOSSA, N85
3817 Strauss Ave., Suite 108
Indian Head, MD 20640-5151

6.4 Service review organizations. The following service contacts are responsible for the assessment of weapon safety and IM characteristics:

Army - (for weapon safety):

SSTM, System Safety US Army DEVCOM Armaments Center
ATTN: FCDD-ACE
BLDG 62
Picatinny Arsenal, NJ 07806-5000

Director
Safety Office US Army Aviation and Missile Command
ATTN: AMAM-SF
BLDG 5300
Redstone Arsenal, AL 35898-5130

Army - (for IM):

U.S. Army DEVCOM
Aviation & Missile Center
Redstone Rd., BLDG 7156
Redstone Arsenal, AL 35898-5000

Navy/Marine Corps - (for weapon safety):

Commanding Officer
Naval Ordnance Safety & Security Activity Farragut Hall
ATTN: NOSSA, N3
3817 Strauss Ave., Suite 108
Indian Head, MD 20640-5151

Navy/Marine Corps - (for IM):

Commanding Officer
Naval Ordnance Safety & Security Activity Farragut Hall
ATTN: NOSSA, N85
3817 Strauss Ave., Suite 108
Indian Head, MD 20640-5151

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Air Force - (for weapon safety):

AFSA/SEWV
9700 G Street
Kirtland AFB, NM 87117-5670

Air Force - (for IM):

ASC/YOX
Eglin AFB, FL 32542-6808

6.5 Tests for HC. The following tests referenced herein have potential application for HC, and some specifics of testing may require approval by service review organizations and the DDESB prior to testing:

a. Safety drop
b. FH
c. SH
d. BI
e. SR
f. FI

Additional tests may be required for HC as defined by TB 700-2.

6.6 Units of measurement and abbreviations. Units of measurement are expressed in metric or SI (Le Système International d'Unités) on the left. The corresponding English equivalent is expressed on the right. Standard abbreviations used throughout this document are as specified in table III.

TABLE III. Units of measurement and abbreviations.

Metric (SI)		**English**	
°C	degrees Celsius	°F	degrees Fahrenheit
G	gram	-	-
Hz	hertz	-	-
J	joule	-	-
mm	millimeters	in	inches
m	meters	ft	feet
MN/m^2	meganewton per square meter	psi	pounds per square inch

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6.7 International standardization agreement implementation. This standard implements the STANAGs listed in table IV.

TABLE IV. Implemented international standardization agreements.

MIL-STD-2105E	NATO Standard
Policy for Introduction and Assessment of Insensitive Munitions (IM)	AOP-39/STANAG-4439
Guidance on the Organisation, Conduct and Reporting of Full-Scale Tests (SRD)	SRD AOP-39.1
Fast Heating test	AOP-4240
Slow Heating test	AOP-4382
Bullet Impact test	AOP-4241
Sympathetic Reaction test	AOP-4396
Fragment Impact test	AOP-4496
Shaped Charge Jet Impact test	AOP-4526
Safety Drop, Munition Test Procedures	STANAG-4375

When changes to, revision, or cancellation of this standard are proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.8 Subject term (key word) listing.

BI

Bullet impact

Fast heating

FH

Fragment impact

IM

Insensitive munitions

SCJ impact test

SH

Shaped charge jet impact

Slow heating

Sympathetic reaction

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – OS
Air Force – 11

Preparing activity:
Navy – OS
(Project SAFT-2021-004)

Review activities:
Army – TE
Navy – AS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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